

Alessandro Demo

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EDUCATION

Polytechnic of Turin

Master's Degree in Computer Engineering, Artificial Intelligence and Data Analytics

Turin, Italy

Nov. 2025 – Present

University of Turin

Bachelor's Degree in Computer Science

Turin, Italy

Sep. 2022 – Nov. 2025

ITIS G.B. Pininfarina

High School Diploma in Electronics and Automation

Moncalieri, Italy

Sep. 2017 – Jul. 2022

EXPERIENCE

Technical Designer Intern

Interactive Fully Electrical Vehicles (I-FEVS)

July 2021

La Loggia, Italy

- Designed the electronic architecture for an e-bike prototype, ensuring reliable embedded system performance.
- Programmed and tested the Portenta H7 board using MicroPython to validate core hardware functionalities.
- Authored comprehensive and precise technical documentation detailing the embedded hardware components.

PROJECTS

Geospatial Data Benchmarking System | Python, GeoPandas, DuckDB, PostgreSQL

May 2025 – Aug. 2025

- Conducted a comparative performance analysis of DuckDB, PostGIS, and GeoPandas for geospatial data processing.
- Designed and implemented benchmarking methodologies in Python to evaluate and document query execution times.
- Managed and processed complex spatial databases utilizing both vector and raster data structures.
- Links:** [Source Code \(GitHub\)](#) | [Read Thesis in Italian \(PDF\)](#)

Sinner Tracker Dashboard | Python, GitHub Actions, RapidAPI, Chart.js

Apr. 2026 – Present

- Engineered a high-performance Progressive Web App (PWA) to track real-time ATP tennis statistics and match data.
- Built automated daily data ingestion pipelines using GitHub Actions to fetch and process data via RapidAPI.
- Visualized complex Head-to-Head (H2H) statistics and ranking progression using interactive charts and responsive UI design.
- Link:** [Source Code \(GitHub\)](#)

Quiz App Engine | JavaScript, HTML/CSS

Jun. 2025 – Apr. 2026

- Developed a responsive, state-managed interactive application using vanilla JavaScript and DOM manipulation.
- Built the core logic and user interface without frontend frameworks to solidify fundamental programming and web proficiencies.
- Link:** [Source Code \(GitHub\)](#)

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript, HTML/CSS

Data Science & AI: Pandas, NumPy, GeoPandas, Jupyter Notebook

Databases: PostgreSQL, PostGIS, DuckDB

Tools & Infrastructure: Git, Linux/Unix, Makefiles, Arduino

Spoken Languages: Italian (Native), English (B2 First Certificate)